Lab Assignment - 2

The objective of this assignment is to investigate how environmental sensor measurements propagate through a feedforward neural network and influence its final output through interactive visualization and experimental analysis. Only forward propagation shall be implemented. Training, gradient computation, optimization, and backpropagation shall not be used.

Experiment: Interactive Visualization of Forward Propagation for Underground Mine Hazard Assessment

1. Develop a Python program using NumPy, Matplotlib, and Matplotlib widgets to simulate an underground mine hazard assessment system. Use methane concentration, oxygen availability, and ambient temperature with default values 0.20, 0.80, and 0.40, respectively.

2. Implement a 3–3–2 fully connected feedforward neural network with three hidden neurons and two output neurons representing Safe Operation and Hazardous Condition. Use the following default parameters:

$$w_1 = \begin{bmatrix} 1.60 & -0.20 & 0.20 \\ -1.20 & 1.80 & -0.30 \\ 0.30 & -0.20 & 1.50 \end{bmatrix} b_1 = [0.20 \quad 0.10 \quad -0.10]$$

$$w_2 = \begin{bmatrix} -1.50 & 1.60 \\ 1.80 & -1.20 \\ -1.30 & 1.50 \end{bmatrix} b_2 = [0.30 \quad -0.20]$$

   In w_1, the rows correspond to three inputs, while the columns correspond to three hidden neurons. In w_2, the rows correspond to the three hidden neurons, while columns are output.

3. Implement Sigmoid, Tanh, and ReLU manually and provide a RadioButtons control to select the activation function during execution. Apply the selected function to both hidden and output layers. Predict Safe Operation when the Safe output is greater than the Hazard output; otherwise, predict Hazardous Condition. Treat both outputs as network scores rather than probabilities.
4. Create one integrated Matplotlib window containing three input sliders, fifteen weight sliders, five bias sliders, the activation-function selection control, a Reset button, a Random Input button, the complete network visualization, and the predicted mine condition. Configure the input sliders from 0.00 to 1.00 with a step size of 0.01, the weight sliders from −2.00 to 2.00 with a step size of 0.10, and the bias sliders from −1.00 to 1.00 with a step size of 0.10. Initialize all sliders using the given default values.

5. Display all input, hidden, and output neurons and their connections in the same window. Show the current inputs, connection weights, neuron biases, weighted sums, hidden activations, output scores, and final prediction up to four decimal places. Use different line colours for positive and negative weights and vary line thickness according to the absolute weight magnitude.
6. Recompute forward propagation and update the existing visualization whenever any input, weight, bias, or activation function changes. The Reset button shall restore all default values and select Sigmoid. The Random Input button shall change only the three environmental inputs while preserving the current weights, biases, and activation function.
7. Using the default network parameters, increase methane concentration from 0.00 to 1.00 while keeping oxygen availability at 0.80 and ambient temperature at 0.40. Record and analyze the changes in hidden activations, output scores, and final prediction.